MAJOR PROJECT

PHARMACY MANAGEMENT SYSTEM

A Centralized Web-Based ERP Platform for Retail Pharmacy Operations

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Centralized Web Architecture | FEFO Automated Inventory | Dynamic GST POS | Smart Patient CRM

INTRODUCTION

From Fragmented Operations to One Centralized Platform

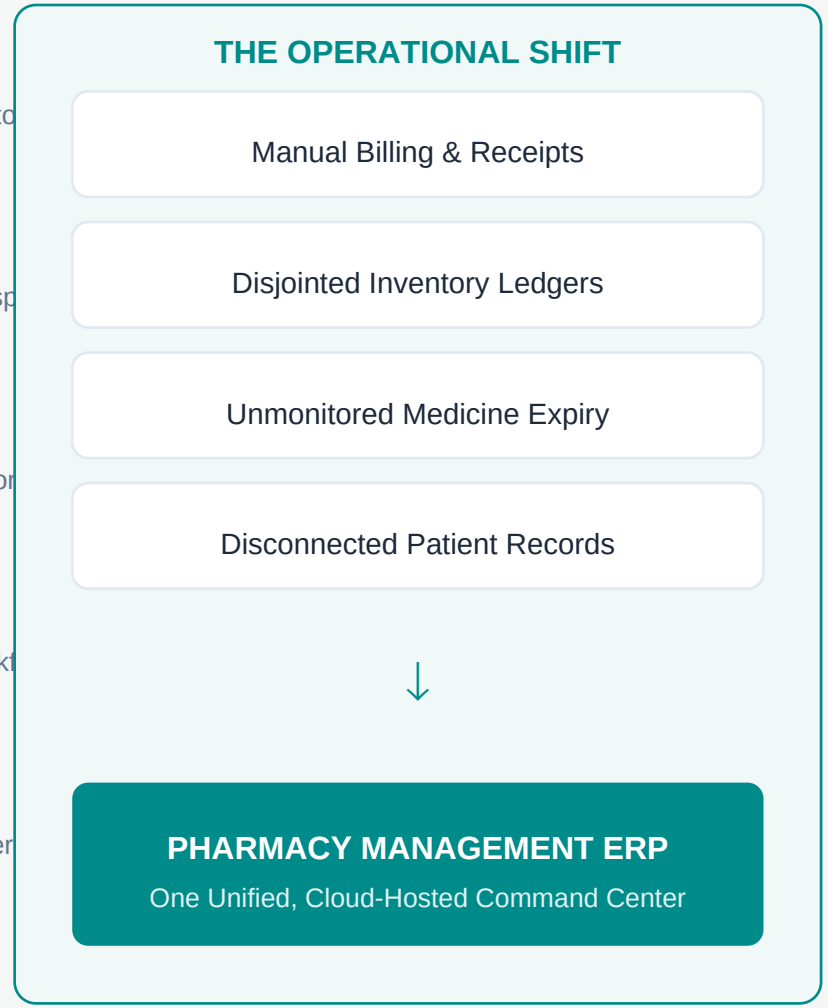
- **Interconnected Retail Nature:**
  Pharmacy retail operations naturally involve heavily interdependent processes — POS customer billing, batch inventc

- **The Fragmented Reality:**
  Traditional retail pharmacies often rely on manual paper registers, standalone billing machines, or isolated desktop sp

- **Operational Bottlenecks:**
  This disconnection makes drug expiry tracking, low-stock reordering, reverse logistics, and overall profit auditing error

- **The Centralized ERP Solution:**
  Pharmacy Management ERP resolves this with a full-stack, cloud-hosted web system that consolidates all retail workf

- **Real-World Operational Alignment:**
  Purpose-built around real daily pharmacy workflows — including rapid barcode scanning, FEFO batch allocation, ther



PROBLEM STATEMENT

Critical Bottlenecks in Traditional Pharmacy Retail

01

Inventory Expiry & Dead Stock

Medicines sit on shelves unnoticed until expiry. Lack of batch-leve

02

Billing Inefficiency & Errors

Manual product lookups and manual pricing calculations slow ds

03

Coarse-Grained Stock Monitoring

Tracking only gross medicine quantities without individual batch numbe

04

Lack of Patient Traceability

Manual customer logging fails to retain prescription history, visitin

05

Blind Financial Visibility

Business owners lack real-time visibility into daily revenue, net pr

06

Absence of Role-Based Security

Cashiers, store managers, and owners share identical access permissio

PROJECT OBJECTIVES

Concrete Engineering Goals of Pharmacy Management ERP

Centralized Web Platform
Build a unified, responsive web ERP accessible across local network and cloud without requi

Smart Patient CRM Directory
Maintain patient profiles, visiting doctors, and cumulative purchase history with seamless AJAX auto-fill

Interactive POS & Barcode Billing
Implement instant medicine lookup via Select2 and physical barcode scanner input for sub-se

Business Intelligence Dashboards
Provide real-time analytics for revenue, net profit, batch expiry timelines, cash/UPI payment distribution

FEFO Batch-Level Inventory
Enforce First-Expire-First-Out logic at the database level to ensure batches expiring earliest a

Reverse Logistics & Returns Desk
Facilitate receipt barcode scanning for customer returns with automated stock shelf replenishment and

Automated Reverse GST Engine
Dynamically compute inclusive GST and tax breakups according to each medicine's registere

Three-Tier Role-Based Security
Enforce strict separation of duties among Administrator (Owner), Store Manager, and Counter Cashier.

Data Flow Diagram — Level 0 (Context Diagram)

COUNTER CASHIER / PHARMACIST

- Barcode / Product queries
- Customer phone & Cart item input
- Receipt & Return verification

→ **Sales Data / Returns** →

← **Thermal Bill / Alerts** ←

STORE MANAGER / ADMIN

- Stock batch & Supplier entries
- RBAC role configuration
- BI analytics & Reorder PO queries

← **New Stock / Roles** ←

→ **BI Reports & POs** →

0.0

PHARMACY MANAGEMENT ERP SYSTEM

Centralized Web & Database Engine

CUSTOMER / PATIENT

- Prescriptions & Contact phone
- Cash / UPI Payment
- Return item with receipt barcode

→ **Prescriptions & Cash** →

← **Invoice & Refunds** ←

SUPPLIER / WHOLESALER

- Medicine batch consignments
- Purchase invoices & MRP data
- Automated reorder CSV orders

← **Invoices & Batches** ←

→ **Purchase Orders (CSV)** →

Context: Central system manages single-point of truth with real-time ACID transactions in SQLite.

SYSTEM MODELING

Data Flow Diagram — Level 1 (Decomposition Diagram)

1.0 Authentication & RBAC Control

Validates login against User/Group models; routes to Admin, Manager, or Cashier UI.

2.0 POS & Reverse GST Billing Engine

Accepts barcode/product queries; calculates GST dynamically; creates Sale transaction.

3.0 FEFO Batch Inventory Dispatch

Sorts stock by expiry_date; deducts sold quantities; triggers Signal replenishment on deletion.

4.0 Reverse Logistics (Returns Desk)

Scans receipt barcode; validates bill items; restocks physical units; logs ReturnSale audit.

5.0 Smart CRM & Customer Directory

Executes AJAX telephone lookup; links customer to doctor, prescriptions, and purchase total.

6.0 Procurement & Reorder Engine

Evaluates batch stocks against minimum_stock_level; exports CSV reorder purchase orders.

7.0 Financial BI & Analytics Engine

Aggregates daily sales, profit margins, GST totals, and payment modes (Cash/UPI).

DATABASE DATA STORES (RELATIONAL REPOSITORY)

D1 Medicine Catalog Table

Attributes: barcode, name, category, manufacturer, pack_size, gst_percentage, min_stock.

D2 Stock Batches Table

Attributes: medicine_id, batch_no, expiry_date, quantity, purchase_price, mrp, supplier_id.

D3 Sale Ledger Table

Attributes: bill_no, stock_id, quantity, total_price, customer_id, payment_method, date.

D4 ReturnSale Audit Table

Attributes: sale_id, stock_id, quantity_returned, refund_amount, reason, restocked, returned_

D5 Customer CRM Table

Attributes: name, phone (unique), last_doctor, cumulative purchase history.

D6 Supplier Directory Table

Attributes: name, contact_person, phone, email, address.

Core Operational Modules of Pharmacy Management ERP

POS & DYNAMIC BILLING

- Barcode scanner input
- Select2 live medicine search
- Real-time cart calculation
- Scannable thermal receipt printing

FEFO BATCH INVENTORY

- Batch-level tracking
- Automatic earliest-expiry allocation
- Visual expiry alert badges
- Automated stock deduction

PATIENT CRM DIRECTORY

- Fast AJAX phone lookup
- Prescribing doctor tracking
- Lifetime purchases & spend
- Returning patient loyalty

FINANCIAL ANALYTICS & BI

- Dynamic GST tax breakdown
- Revenue & profit margins
- Best-selling medicine ranking
- CSV financial export

REVERSE LOGISTICS DESK

- Receipt barcode returns
- Return reason logging
- Automatic shelf restocking
- Customer refund computation

SECURITY & RBAC

- Admin, Manager, Cashier roles
- View-level access restrictions
- Sensitive cost masking
- Auto session timeout on close

SYSTEM ARCHITECTURE

Three-Tier Modular Web Architecture

Presentation Layer (Client)

HTML5 • CSS3 • Bootstrap 5.3 • Vanilla JavaScript (ES6) • AJAX • Select2 • JsBarcode • FontAwesome

Application Layer (Business Logic)

Python 3.14 • Django 6.0.4 • Django Views & Middleware • Django ORM • FEFO Engine • Signals • RBAC Engine

Data Layer (Persistent Storage)

SQLite Database (Production Deployment) • Relational Models (Medicine, Stock, Sale, ReturnSale, Customer, Supplier)

Cloud Infrastructure & Security

PythonAnywhere Cloud PaaS • Django WSGI • Session Timeout Security • python-dotenv Secret Vault

TOOLS & TECHNOLOGIES

Full Implementation Technology Stack

Programming Language

Python 3.14 (High-efficiency backend execution)

Backend Framework

Django 6.0.4 (MVC/MVT modular framework)

Database Engine

SQLite / Relational Schema (Zero-config, ACID-compliant)

Object-Relational Mapper

Django ORM (Safe parameterized queries & transactions)

UI Presentation

HTML5, CSS3, Bootstrap 5.3 (Responsive desktop/tablet UI)

Client-Side Scripting

Vanilla JavaScript (ES6) (Fast DOM & event handling)

Asynchronous Operations

AJAX (Background patient search & master additions)

Search Autocomplete

Select2 (Fuzzy fast medicine lookup in POS)

Barcode Generation

JsBarcode (Thermal bill scannable CODE128 generation)

Data Visualization

Chart.js (Interactive financial & revenue line charts)

Environment Security

python-dotenv (Vault isolation for SECRET_KEY)

Cloud Deployment

PythonAnywhere & Django WSGI (Public web access)

Core Algorithms & Implementation Highlights

Automated FEFO Dispatch Algorithm

Instead of arbitrary manual batch picking, queries evaluate all active

Dynamic Reverse GST Calculation

Since medicine MRPs are legally tax-inclusive in India, the billing engi

Event-Driven Django Signals

Implemented post_delete receiver signal on Sale transactions to trigger

Asynchronous Patient Auto-Fill

Typing customer phone trigger debounced AJAX queries to /api/get_

Multi-Tier Role-Based Security

Custom middleware and template permission checks verify request

Enterprise POS Session Hardening

Configured SESSION_EXPIRE_AT_BROWSER_CLOSE = True and SESS

Relational Entity Schema & Data Relationships

Medicine (Product Master)

- barcode (Unique)
- name, category
- manufacturer
- gst_percentage
- minimum_stock_level
- pack_size
- @property total_stock

Stock (Batch Inventory)

- batch_no
- expiry_date (Date)
- quantity (Int)
- purchase_price (Float)
- mrp (Float)
- FK -> Medicine (CASCADE)
- FK -> Supplier (SET_NULL)

Sale (Transaction Ledger)

- bill_no (Indexed)
- quantity (Int)
- total_price (Float)
- payment_method (Cash/UPI)
- date (DateTimeField)
- FK -> Stock (CASCADE)
- FK -> Customer (SET_NULL)

ReturnSale (Audit Trail)

- quantity_returned (Int)
- refund_amount (Float)
- reason (WRONG/DEFECT/EXCESS)
- restocked (Boolean)
- returned_at (DateTime)
- FK -> Sale (CASCADE)
- FK -> Stock (CASCADE)

Customer (Smart CRM)

- name (Char)
- phone (Unique 15)
- last_doctor (Char)
- related: purchase_history
- computed: total_spent

Supplier (Wholesaler)

- name (Char 200)
- contact_person
- phone, email
- address (Text)
- related: stock_set

PROJECT PROGRESSION

Phased Development Milestones (Major Project Build)

P1

Phase 1: Requirements & Modeling

Analysis of pharmaceutical retail workflows, GST compliance rules, FEFO logic, and SQLite relational database schema.

P2

Phase 2: Core Backend Engine

Implemented Django models, Django ORM relationships, management commands, and database migrations.

P3

Phase 3: Interactive POS Billing

Developed quick-checkout billing interface with dynamic Select2 searching, barcode reader input, and thermal receipt generator.

P4

Phase 4: Inventory & FEFO Engine

Built batch management, automatic earliest-expiry deduction, low-stock reorder CSV export, and batch entry forms.

P5

Phase 5: Reverse Logistics & Returns

Engineered returns desk with bill barcode verification, refund calculation, and automated stock shelf restoration.

P6

Phase 6: CRM & Business Intelligence

Integrated asynchronous patient lookup, customer purchase history directory, and real-time Chart.js KPI analytics.

P7

Phase 7: Security & Cloud Deployment

Enforced RBAC permissions, environment variable secret isolation, session hardening, and cloud deployment on PythonAnywhere.

APPLICATIONS & USE CASES

Practical Deployment Scenarios for Pharmacy ERP

INDEPENDENT RETAIL PHARMACIES

Single-location medical stores that require quick barcode billing, automated FEFO expir

HOSPITAL OUTPATIENT DISPENSARIES

High-volume hospital pharmacies that demand high-speed customer lookup, doctor-prescription

CHAIN & MULTI-COUNTER STORES

Multi-terminal medical retail stores with designated cashier and store manager roles, pre

RURAL HEALTHCARE CENTERS

Low-overhead deployment capable of running on modest hardware and low bandwidth with relia

FUTURE SCOPE

Phase 2 Technological Roadmap

AI & MACHINE LEARNING

- OCR-based doctor prescription parsing
- Predictive medicine stock replenishment
- Seasonal disease outbreak demand forecasting

MOBILE & REST API ECOSYSTEM

- Django REST Framework (DRF) API endpoints
- Mobile Android/iOS apps for store owners
- Bluetooth mobile thermal printer integration

COMMUNICATION AUTOMATION

- WhatsApp Business Cloud API integration
- Automated SMS digital receipt delivery
- Automated low-stock email alerts to suppliers

ENTERPRISE SCALABILITY & DEVOPS

- Seamless migration from SQLite to PostgreSQL
- Docker containerization & microservice scaling
- Automated CI/CD pipeline deployment

CONCLUSION & SUMMARY

Transforming Retail Pharmacy into an Automated Digital Enterprise

✔ **Centralized Web Efficiency:**

Successfully eliminated disjointed paper ledgers and fragmented software by consolidating POS, inventory, CRM, and analytics into one web platform.

✔ **Elimination of Expiry Losses:**

Enforced automated FEFO batch allocation, preventing older stock from expiring unnoticed on shelves.

✔ **Compliance & Billing Speed:**

Implemented dynamic reverse GST calculations and barcode scanning, enabling rapid, error-free checkout counter operations.

✔ **Auditability & Reverse Logistics:**

Delivered barcode-verified customer returns desk with automatic shelf inventory recovery and ReturnSale audit logs.

✔ **Academic & Practical Success:**

Fully implemented, tested, and cloud-deployed on PythonAnywhere with complete role-based security, ready for real-world deployment.

THANK YOU | QUESTIONS & DISCUSSION